

CORVALLIS, OR, USA

WMO: 726945

Lat: 44.500N

Lon: 123.283W

Elev: 76

StdP: 100.41

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24202

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-4.0	-2.5	-8.1	1.9	-2.0	-5.9	2.3	-0.5	11.0	9.3	10.0	8.8	0.9	300	0.533

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.0	33.8	19.5	32.1	19.1	29.7	18.1	20.6	31.5	19.6	30.5	18.7	28.5	4.2	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.3	11.7	24.2	15.2	10.9	23.6	14.1	10.1	22.2	59.2	31.4	55.8	30.3	53.0	28.5	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	8.0	7.2	-7.7	38.3	3.6	1.9	-10.2	39.6	-12.3	40.7	-14.3	41.8	-16.9	43.2		
(5)			-8.2	22.6	3.4	1.3	-10.7	23.5	-12.7	24.3	-14.7	25.1	-17.2	26.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.4	5.5	6.6	8.7	11.0	14.6	17.3	20.8	20.6	17.6	12.5	8.0	5.2		
	DBStd	6.32	3.53	3.09	2.85	2.89	3.24	3.10	2.76	2.56	2.89	3.05	3.62	3.58		
	HDD10.0	571	144	100	58	21	3	0	0	0	0	11	81	153		
	HDD18.3	2383	398	328	297	221	123	55	8	8	47	181	311	406		
	CDD10.0	1449	4	6	19	51	145	219	336	328	228	89	19	4		
	CDD18.3	221	0	0	0	1	7	24	85	78	25	1	0	0		
	CDH23.3	3192	0	0	1	27	172	398	1085	1006	465	38	0	0		
	CDH26.7	1302	0	0	0	4	51	153	482	434	171	7	0	0		
(14) Wind	WSAvg	2.8	2.3	2.5	2.8	2.7	2.8	3.0	3.7	3.4	2.8	2.4	2.4	2.5		
(15) Precipitation	PrecAvg	1367	225	156	152	96	65	38	7	13	39	114	208	255		
	PrecMax	2288	512	436	286	198	140	82	26	51	179	299	572	563		
	PrecMin	900	57	8	27	37	0	5	0	0	3	15	27	96		
	PrecStd	315	105	96	63	46	35	18	7	14	37	61	108	112		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.4	17.2	22.1	27.2	31.2	35.0	37.6	37.1	33.6	27.4	18.7	15.1		
		MCWB	12.8	11.3	13.8	16.6	19.0	19.6	21.2	20.0	18.0	16.3	16.0	12.7		
	2%	DB	13.5	14.5	18.5	22.7	27.7	31.0	33.8	33.0	31.1	23.7	16.3	13.0		
		MCWB	11.1	10.4	12.0	14.3	16.9	18.6	19.9	19.4	17.8	14.7	13.0	10.8		
	5%	DB	12.2	12.9	16.3	19.9	24.8	27.6	32.0	31.2	28.6	21.3	14.6	12.1		
		MCWB	10.1	9.7	10.9	12.9	15.7	17.3	19.4	18.9	17.3	14.2	11.8	10.4		
	10%	DB	11.0	12.1	13.9	17.3	22.2	25.0	29.0	28.8	26.1	18.7	13.0	10.8		
		MCWB	9.1	9.2	9.9	11.5	14.6	16.0	18.3	18.1	16.3	13.5	10.7	9.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.3	12.9	14.7	17.2	19.7	21.1	22.4	21.5	20.2	18.0	16.3	13.2		
		MCDB	15.2	15.2	20.1	26.1	29.8	32.4	34.8	33.4	29.6	24.1	18.2	14.8		
	2%	WB	11.4	11.4	12.9	15.0	17.7	19.1	20.7	20.3	18.8	16.3	14.0	11.5		
		MCDB	13.1	13.7	17.0	21.8	26.5	29.2	31.8	31.4	28.2	21.3	16.1	12.9		
	5%	WB	10.2	10.2	11.6	13.3	16.2	17.8	19.7	19.4	17.8	15.0	12.4	10.2		
		MCDB	11.8	12.6	15.4	19.0	23.9	27.0	31.1	29.9	26.8	19.2	14.2	11.8		
	10%	WB	9.0	9.1	10.5	11.8	14.9	16.5	18.6	18.4	16.7	14.0	11.2	8.7		
		MCDB	10.6	11.4	13.7	16.5	21.5	24.2	28.6	28.0	24.7	17.7	13.0	10.1		
(35) Mean Daily Temperature Range	MDBR		6.3	8.3	9.8	11.3	12.7	13.6	16.0	16.5	15.7	12.3	7.8	6.0		
	5% DB	MCDBR	7.3	10.6	13.7	17.3	18.7	18.9	20.3	20.8	21.5	17.6	8.8	6.3		
		MCWBR	5.5	7.2	8.6	10.0	9.9	9.0	8.9	9.1	10.2	10.0	6.4	5.2		
	5% WB	MCDBR	6.8	8.4	12.2	15.5	16.8	17.3	19.0	18.7	18.3	13.2	7.0	6.0		
		MCWBR	5.5	6.2	8.1	9.3	9.3	8.6	8.8	8.6	9.2	8.1	5.5	5.1		
(40) Clear-Sky Solar Irradiance	taub		0.317	0.317	0.324	0.357	0.364	0.358	0.351	0.356	0.345	0.327	0.324	0.327		
	taud		2.534	2.542	2.529	2.398	2.396	2.419	2.457	2.464	2.486	2.548	2.522	2.468		
	Ebn at Noon		810	879	917	906	905	909	912	896	880	854	791	750		
	Edh at Noon		67	79	90	112	116	113	108	103	93	77	67	66		
(44) All-Sky Solar Radiation	RadAvg		1.31	2.11	3.10	4.46	5.67	6.29	7.17	6.14	4.61	2.72	1.48	1.06		
	RadStd		0.19	0.27	0.37	0.37	0.54	0.56	0.43	0.38	0.38	0.31	0.19	0.15		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+17

Nomenclature: See separate page